SERVICE - ORIENTED SOFTWARE ENGINEERING

DT0203

Introduction & Preliminary Concepts

Towards Data as a Service

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What is a Data?

1st Definition of Data

« A collection of facts, information and statistics that can be analysed to develop new knowledge »

From Open Data in a Day by Dave Tarrant (Open Data Institute - ODI)

Open Data Institute (ODI) is a non-profit private company, based in the United Kingdom, whose mission is to connect, equip and inspire people around the world to innovate with data

2nd Definition of Data

« A collection of numbers assigned as values to quantitative variables and/or characters assigned as values to qualitative variables »

From Open Data in a Day by Dave Tarrant (Open Data Institute - ODI)

3rd Definition of Data

« The lowest level of abstraction from which information and then knowledge are derived »

From Open Data in a Day by Dave Tarrant (Open Data Institute - ODI)

Definition of Data

By referring the 3rd definition of Data and concerning this course we consider the following

Data is the lowest level of abstraction from which information and then knowledge and wisdom are derived

Definition of Data

By referring the 3rd definition of Data and concerning this course we consider the following

Data consists of raw facts or symbols that have **no interpretation, context, or meaning by themselves**.

Abstraction means simplifying reality by focusing on important aspects and ignoring details.

Data is the lowest level of abstraction from which information and then knowledge and wisdom are derived

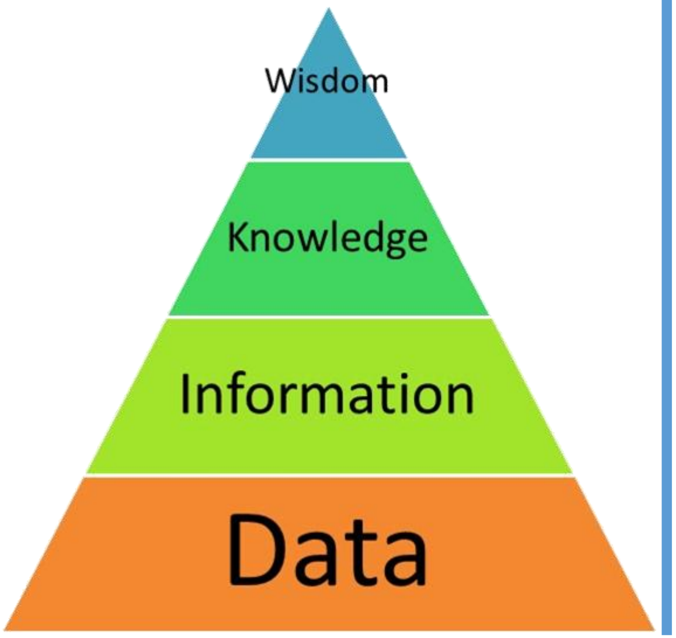
From Data to Wisdom

Data, **I**nformation, **K**nowledge and **W**isdom (**DIKW**) are in hierarchical relationship

Pronounced as: Dee - I - Key - W Pyramid or "Dik-W" Pyramid (some pronounce it like "Dick-W")

DIKW pyramid is a popular model for referring to the representation of the relationships between data, information, knowledge and wisdom

- also variously known as the DIKW hierarchy, Wisdom Hierarchy, Knowledge Hierarchy, Information Hierarchy, and the Knowledge Pyramid



Each step up the pyramid answers questions about and adds value to the initial data

« Typically, information is defined in terms of data, knowledge in terms of information, and wisdom in terms of knowledge » [1]

[1] Rowley, Jennifer. (2007). "The wisdom hierarchy: representations of the DIKW hierarchy", *Journal of Information Science*, 33(2), pp. 163–180.

DIKW Pyramid

Data is defined as symbols that represent properties of objects, events and their environment and can be the product of observation

Data are not useable (i.e., irrelevant/meaningless) until we cannot associate additional information with them

- Raw values or measurements, e.g., the string red or the number 28 (raw data) without meaning and purpose can mean little

DIKW Pyramid

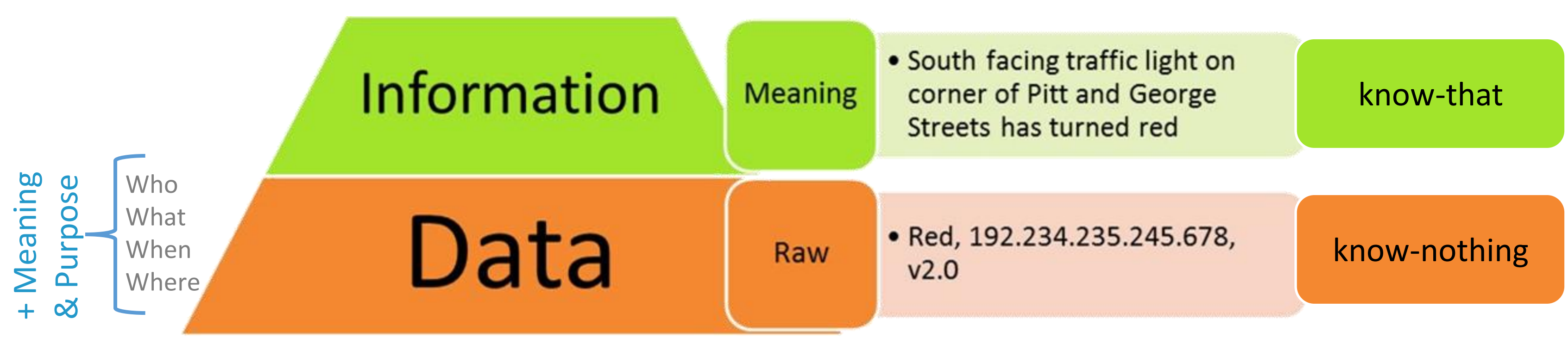
Data + Meaning and Purpose = Information

Context is supplemental data that provides the meaning to the original data set

Valuable information can be derived by answering to questions that begin with **WHO**, **WHAT**, **WHEN**, **WHERE**, etc. (not HOW!)

- e.g., South facing traffic light on corner of Pitt and George Streets has turned red

  (Note that both data and information are often objects in Nature)

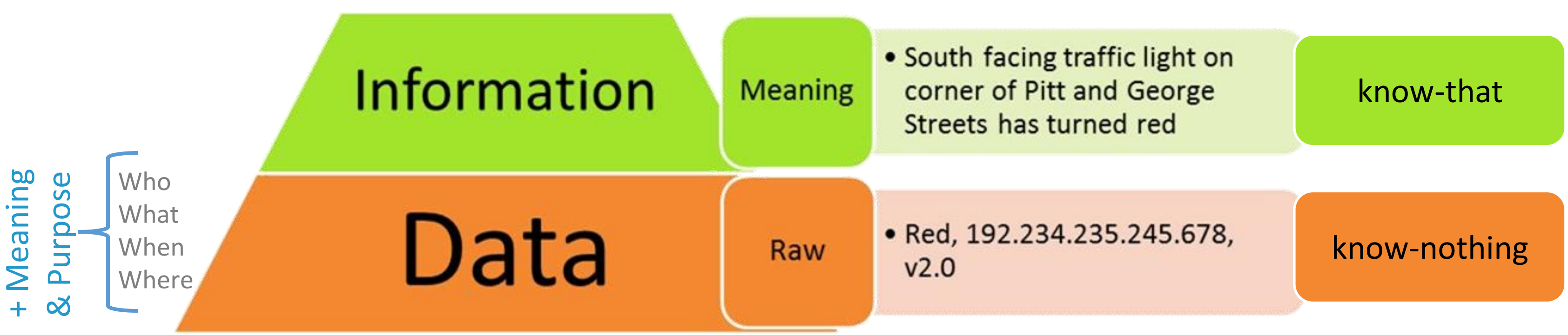


DIKW Pyramid

Information is the meaning of collected data, possibly, cleaned of errors and further processed to easier measure, visualize and analyze

Data processing can involve different operations such as

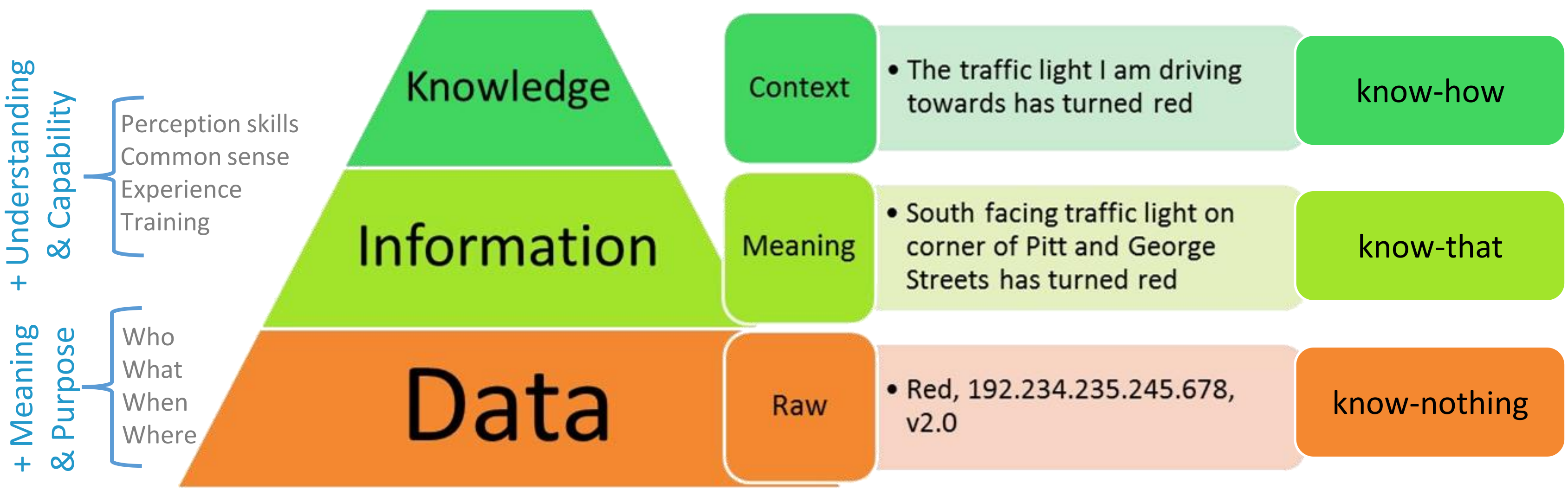
- aggregation: combining different sets of data
- validation: ensuring that the collected data is relevant and accurate
- . . .



DIKW Pyramid

Information + understanding & capabilty = knowledge

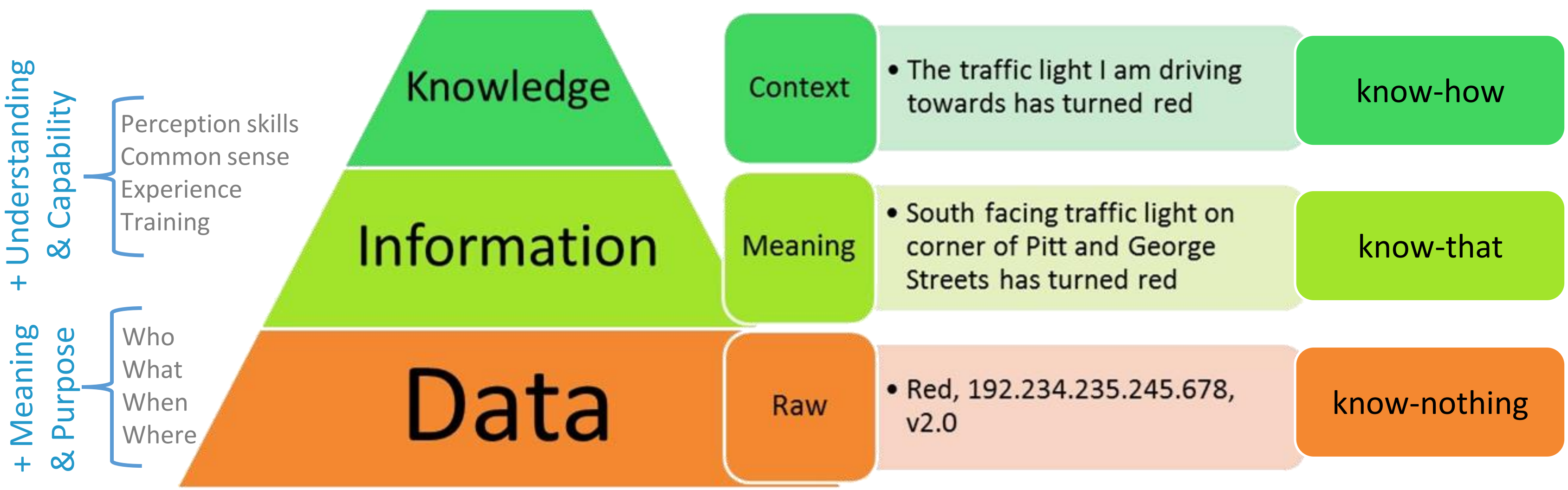
When we do not consider information as a description of collected facts, but understand **HOW** to use/apply the information to achieve our goals, we turn the information into knowledge



DIKW Pyramid

Thus, knowledge can be derived by answering to questions that begin with HOW

- e.g., HOW can we apply the information to achieve our goal?
- e.g., HOW is the information, derived from the collected data, relevant to our goals?
- e.g., HOW are the pieces of this information connected to other pieces to add more meaning and value?
- . . .

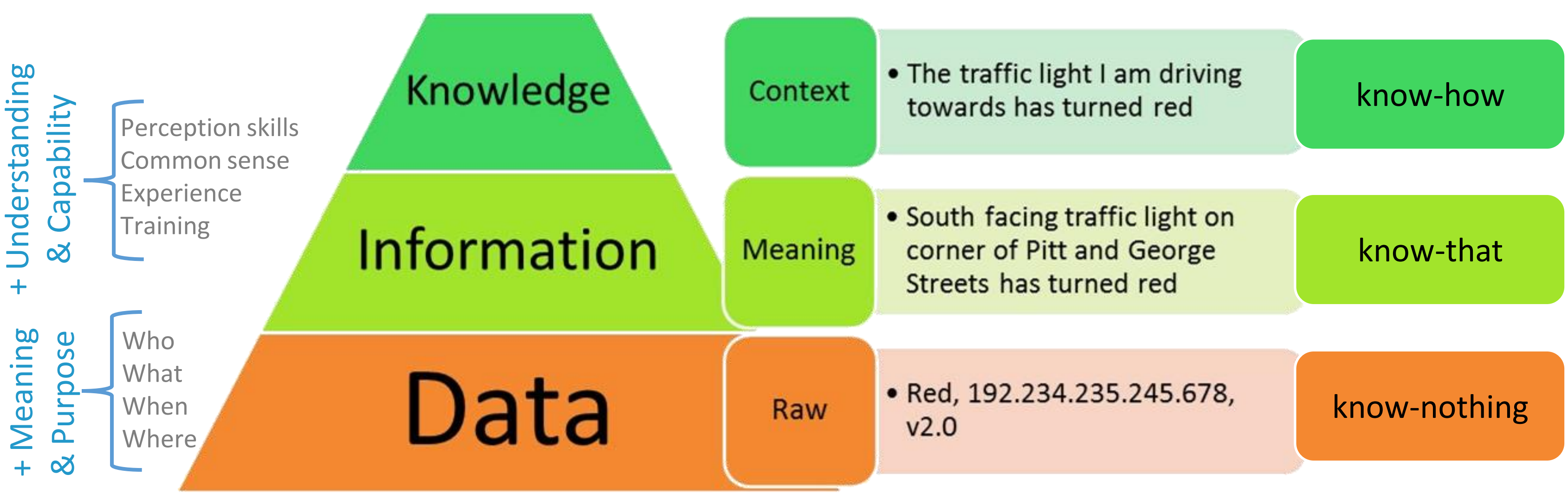


DIKW Pyramid

When we discover relationships that are not explicitly stated as information, we get deeper insights that take us higher up the DIKW pyramid

Notice that with the knowledge, we are incorporating subjective expertise as different people have different experiences to shape their understanding of the information

Thus, pay attention; we are Data Analysts, Computer Scientists! → next slide



Knowledge acquisition and interpretation

The role of subjective expertise in shaping how people understand and apply information is crucial

Subjectivity in Information Interpretation

Knowledge is not always objective; different individuals bring their own **perspectives, biases, and experiences** when interpreting information

For example, a data analyst may focus on statistical significance in a dataset, while a computer scientist may prioritize algorithm efficiency when working with the same data

The Influence of Experience

People’s past experiences shape their understanding

A senior data analyst might immediately recognize certain data patterns based on years of exposure, whereas a beginner might interpret the same information differently

The Importance of Critical Thinking

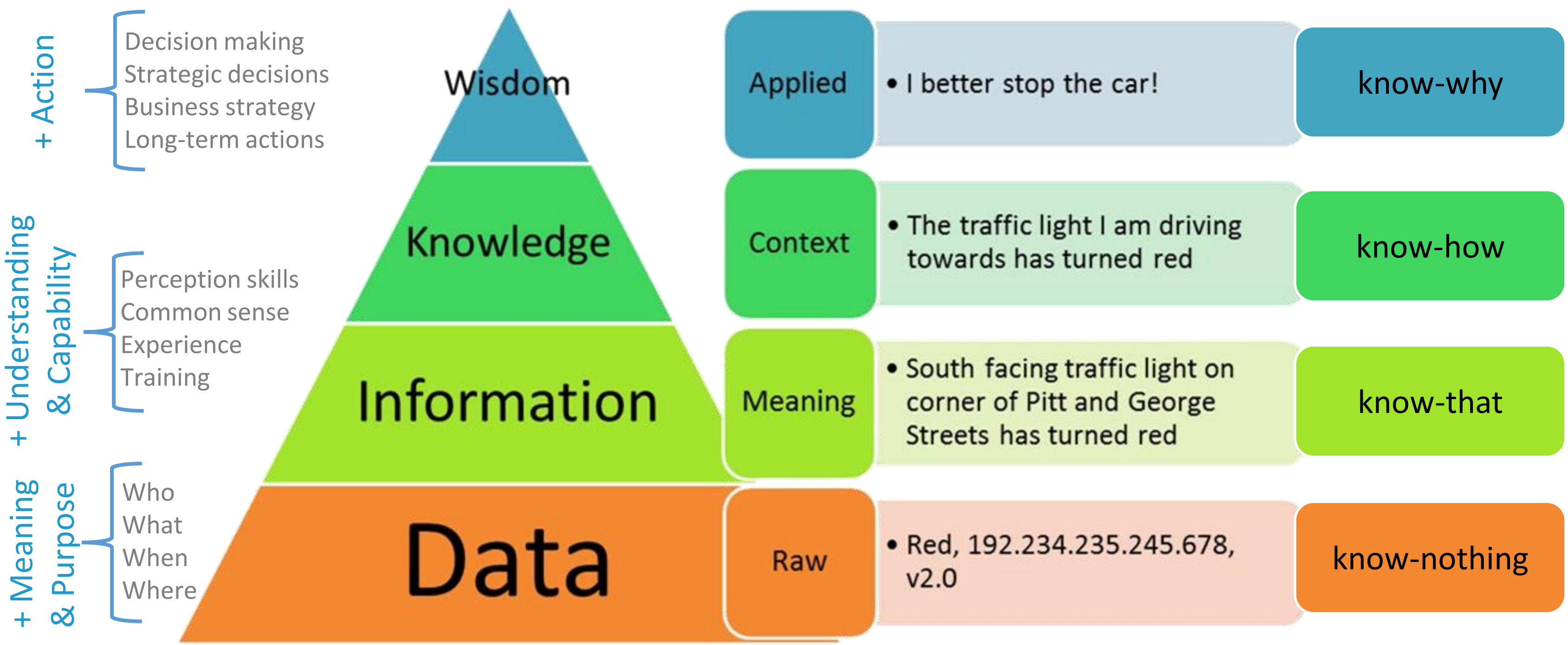
Being aware of this subjectivity is crucial for professionals like Data Analysts and Computer Scientists who rely on data-driven decision-making

While expertise is valuable, critical thinking is necessary to separate facts from personal biases when analyzing information

DIKW Pyramid

When we use the knowledge and insights gained from the information to take proactive decisions, we can say that we have reached the wisdom

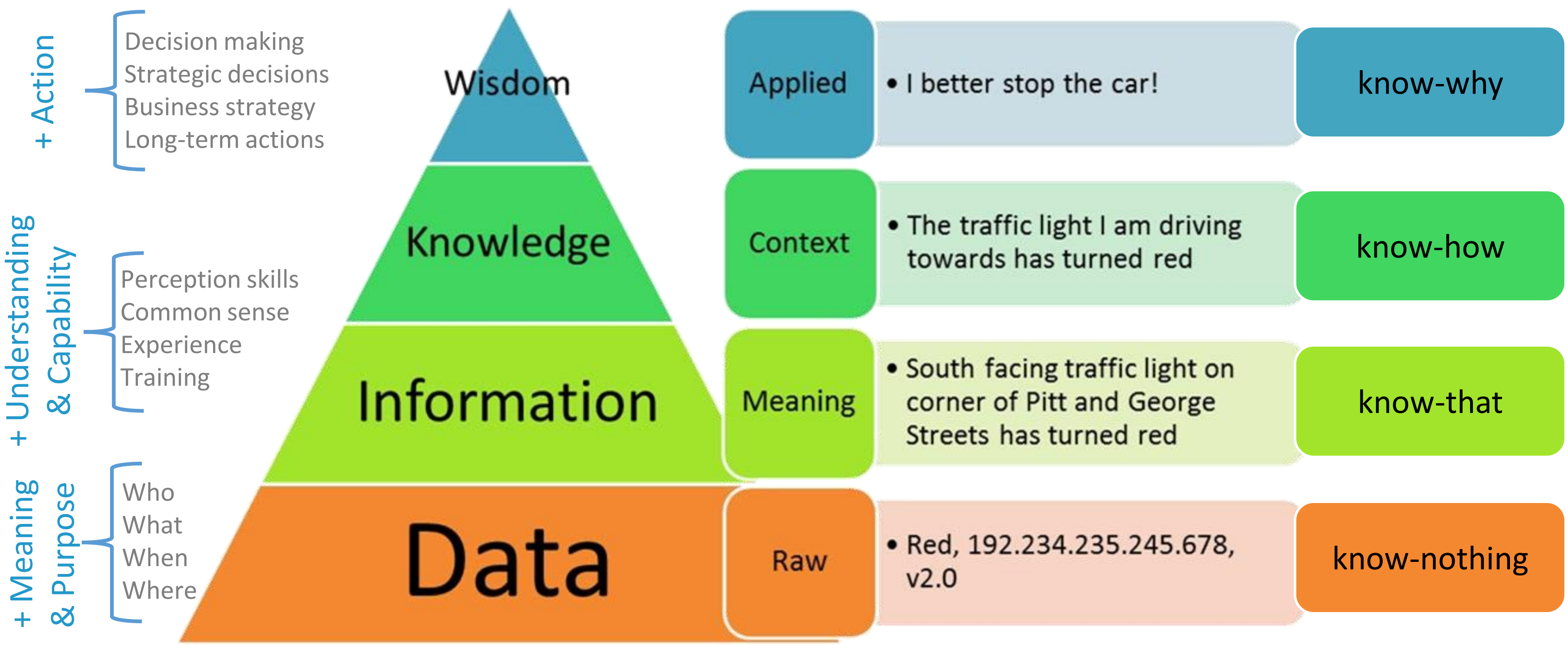
Wisdom is the ability to increase the effectiveness



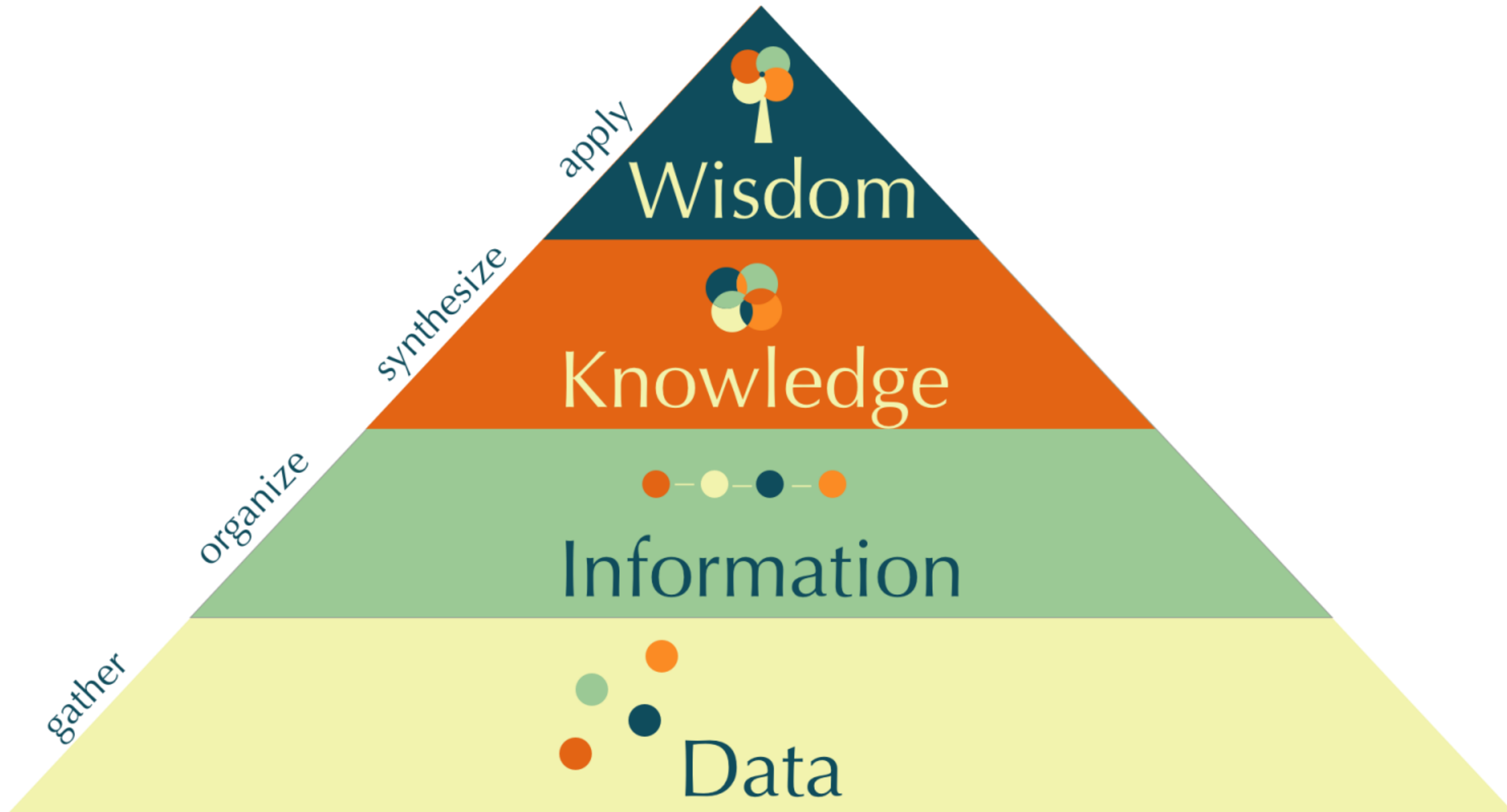
DIKW Pyramid

Wisdom is knowledge applied in action, e.g., “I better stop the car!”

Wisdom is also subjective



Many other representations



Enterprises/organizations *VS* DIKW Pyramid

How enterprises/organizations move up the Knowledge Pyramid?

One easy and fast way for enterprises to take the steps from data to information to knowledge and wisdom is to use Semantic Technologies such as Linked Data and Semantic Graph Databases

These technologies can create links between disparate and heterogeneous data and infer new knowledge out of existing facts

Armed with this new knowledge, enterprises can climb up the mountain of wisdom and gain a competitive advantage by supporting their business decisions with data-driven analytics

Now let's see some more complete
and concrete examples ☺

1. Analyzing Customer Behavior in an E-Commerce Store

Data (Raw Facts)

- The system collects unprocessed, raw **transactional** and **behavioral data**:
  - **Customer ID:** 12345
  - **Product:** Wireless Mouse
  - **Price:** $50
  - **Purchase Date:** March 1, 2025
  - **Page Views Before Purchase:** 5
  - **Cart Abandonment Rate:** 40%

- This is just **isolated data points** without precise meaning and purpose

1. Analyzing Customer Behavior in an E-Commerce Store

Information (Processed Data with Meaning)

- The **data is structured** and **summarized** into meaningful insights:
  - The **average customer** views a product **4-6 times before purchasing**
  - Customers who abandon their carts **tend to leave without returning within 24 hours**
  - The wireless mouse has a **high conversion rate (10%)** compared to similar products
- This gives a **clearer picture** (**understanding**) of customer behavior

1. Analyzing Customer Behavior in an E-Commerce Store

Knowledge (Actionable Insights)

- By **analyzing trends** and **correlations**, we gain **predictive knowledge**:
  - Customers **who view a product 5+ times** but do **not buy immediately** are more likely to abandon their cart
  - Sending a **personalized email with a limited-time discount** within **6 hours** of cart abandonment **boosts conversions by 15%**
  - Customers **prefer products with high ratings**, so emphasizing **customer reviews** on product pages increases engagement

- These insights allow the business to **optimize marketing efforts** and **sales strategies**

1. Analyzing Customer Behavior in an E-Commerce Store

Wisdom (Strategic Decision-Making)

- **Applying experience** and **business strategy** to make **long-term**, **high-level decisions**:
  - Instead of just targeting **abandoning customers with discounts**, the company **redesigns the product page** to improve conversions **before abandonment happens**
  - The **email marketing team** shifts from generic emails to **AI-driven personalization**, offering targeted promotions based on browsing behavior
  - **AI chatbots** are introduced to provide real-time assistance for customers hesitating at checkout
  - Based on insights, **pricing adjustments** are tested:
    - Slightly increasing the price but bundling **“Wireless Mouse + Mouse Pad”** increases overall revenue
    - A **subscription-based model** for frequent buyers is introduced

1. Analyzing Customer Behavior in an E-Commerce Store

Summary

Stage	Meaning	Example in E-Commerce
Data	Raw numbers and records	Customer 12345 viewed a product 5 times and bought it for $50
Information	Processed data with meaning	Customers who view a product 5+ times are more likely to abandon carts
Knowledge	Actionable insights	Sending a discount email within 6 hours reduces cart abandonment by 15%
Wisdom	High-level, strategic decisions	Improve product pages, use AI for personalization, introduce a subscription model

1. Analyzing Customer Behavior in an E-Commerce Store

Final Takeaway

- **Data Analysts** extract and interpret raw data
- **Computer Scientists & AI Experts** develop personalized recommendation systems
- **Marketing & Business Leaders** apply **wisdom** to drive strategy, ensuring long-term success

- This approach moves **beyond just reacting to customer behavior**, it **anticipates** and **influences** future behavior for sustainable growth

2. Heart Rate Monitoring with a Smartwatch

A smartwatch tracks a user’s heart rate and provides actionable insights over time

Data (Raw Facts)

- The smartwatch collects raw **sensor readings** without context:
  - Heart Rate at 10 AM: **88 bpm**
  - Heart Rate at 2 PM: **110 bpm**
  - Heart Rate at 6 PM: **95 bpm**

- This is just numbers, **no meaning yet**

2. Heart Rate Monitoring with a Smartwatch

Information (Processed Data with Meaning)

- The smartwatch **organizes** and **summarizes** the data:
  - The user's **average daily heart rate** is **94 bpm**
  - Heart rate tends to **spike after workouts** but returns to normal within an hour
- Now, the data has **structure** and **relevance**, but we still need interpretation

2. Heart Rate Monitoring with a Smartwatch

Knowledge (Actionable Insights)

- By analyzing **patterns** and **trends**, the smartwatch provides insights:
  - The heart rate **exceeds 120 bpm after running for 10 minutes**
  - The user's resting heart rate is **consistently above 90 bpm**, which may indicate potential cardiovascular issues

- This knowledge can help the user **understand their health condition** better

2. Heart Rate Monitoring with a Smartwatch

Wisdom (Decision-Making and Long-term Actions)

- Based on knowledge, the user (or a doctor) makes a decision:
  - Since a resting heart rate above 90 bpm can be a sign of **hypertension**, the smartwatch suggests **consulting a doctor**
  - The user **adjusts their workout intensity** and adopts a healthier lifestyle
  - The smartwatch recommends **meditation or breathing exercises** to maintain a healthy heart rate

- Turning Wisdom into Knowledge involves **experience** and **expertise**, then, **judgment** is required to make the best decisions

2. Heart Rate Monitoring with a Smartwatch

Summary

Stage	**Meaning**	**Example from Smartwatch**
Data	Raw numbers	Heart rate readings (88 bpm, 110 bpm, 95 bpm)
Information	Organized data with meaning	Average daily heart rate is 94 bpm
Knowledge	Actionable insights	Resting heart rate is consistently high, may indicate health concerns
Wisdom	Decision-making	Consult a doctor, adjust workout routine, practice meditation

2. Heart Rate Monitoring with a Smartwatch

Final Takeaway

- **Real-World Application**
- **Data Analysts** may work with large datasets to extract meaningful patterns
- **Computer Scientists** may design algorithms to automate insights (e.g., AI-powered health alerts)
- **Doctors and users** apply **wisdom** to make the best decisions for long-term health

What is Open Data?

The concept of open data itself is recent but is not new, gaining popularity with the rise of the Internet and World Wide Web and, especially, with the launch of open-data government initiatives such as

- data.gov (United States Government - March 2009)
- data.gov.uk (United Kingdom Government - January 2010)
- data.gov.in (India Government - October 2012)
- . . .

Among the several formal definitions of Open Data given by different organizations, we consider:

- Open Definition (by the Open Knowledge Foundation)
- Open Data Institute

Open Knowledge Foundation

Open Knowledge Foundation

- previously known as Open Knowledge International (OKI)
- founded in 2004
- is a global non-profit organisation focused on realising open data's value to society by helping civil society groups access and use data to take action on social problems

Open Definition is a project of Open Knowledge Foundation

- created in 2005
- sets out principles that define “openness” in relation to data and content
- ensures quality and encourages compatibility between different pools of open material

https://opendefinition.org/

Open Definition Project

Definition of Open v1.0

« A piece of data or content is open if anyone is free to use, reuse, and redistribute it - subject only, at most, to the requirement to attribute and/or share-alike »

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Retired August 2014

Open Definition Project

Definition of Open v2.0

« Open means anyone can freely access, use, modify, and share for any purpose - subject, at most, to requirements that preserve provenance and openness »

https://opendefinition.org/

Open Definition Project

Definition of Open v2.0

« Open means anyone can freely access, use, modify, and share for any purpose - subject, at most, to requirements that preserve provenance and openness »

Definition of Open Data

« Open data is data that can be freely used, shared and built on by anyone, anywhere, for any purpose »

Three important principles behind this definition, which are why Open Data is so powerful:

- Availability and Access: that people can get the data
- Re-use and Redistribution: that people can reuse and share the data
- Universal Participation: that anyone can use the data

Open Definition Project

The definition of Open is important for Data cause is acts as a standard ensuring:

- Quality: open data should mean the freedom for anyone to access, modify and share that data
- Compatibility: without an agreed definition it becomes impossible to know if your "open" is the same as my "open"
- Simplicity: simplicity and ease of use of open data

Open Data Institute

- founded in 2012 by the inventor of the web **Sir Tim Berners-Lee** and artificial intelligence expert **Sir Nigel Shadbolt**
- independent, non-profit, non-partisan company that shows the value of open data, and to advocate for the innovative use of open data to effect positive change across the globe

Definition of Open Data v1.0

« Open data is information that is available for anyone to use, for any purpose, at no cost »

https://theodi.org

Open Data Institute

Definition of Open Data v1.0

« Open data is information that is available for anyone to use, for any purpose, at no cost »

Retired November 2014

Definition of Open Data v2.0

« Open data is data that anyone can access, use and share »

https://theodi.org

What Makes Data Open?

There are 2 important elements to openness

LEGALLY OPEN	TECHNICALLY OPEN
Data must be placed in the public domain or under liberal terms of use with minimal restrictions	Data must be published in electronic formats that are machine readable and non-proprietary, so that anyone can access and use the data using common, freely available software tools
Legal openness is usually provided by applying an appropriate (open) license which allows for free access to and reuse of the data	Data must also be publicly available and accessible on a public server, without password or firewall restrictions To make Open Data easier to find, most organizations create and manage Open Data catalogs

http://opendatatoolkit.worldbank.org/en/essentials.html

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License for openly access and reuse

p... by applying an appropriate (open) license which allows for openly access to and reuse of the data

TECHNICALLY OPEN

Data must be published in electronic formats that are machine readable and non-proprietary, so that anyone can access and use the data using comm... freely ... tools

No tables in PDF documents, no screenshots

Dat... ...ely available and accessible on a pub...c server, without password or firewall restrictions

To make Open Data easier to find, most organizations create and manage Open Data catalogs

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Unlocking value from Open Data - 1

- Open data has the potential to help grow economies, transform societies and protect the environment
  - A startup uses open transport data to build a traffic app that reduces congestion and pollution.

- Open data stimulates innovation by removing barriers to access, use and shareability of data
  - Developers create new health apps because government health statistics are freely available online.

- Open data is a natural resource of the digital age, but unlike other natural resources like coal or diamonds, it can be used by everyone at the same time
  - Thousands of researchers worldwide use the same open climate dataset simultaneously to study global warming.

- Open data literacy is a crucial skill for any company that wants to adapt to a changing world and to take advantage of this new boom in resources
  - A company trains its employees to analyze open market data to identify new business opportunities.

Unlocking value from Open Data - 2

Innovation and growth in businesses

- open data is supporting innovation and growth by revealing opportunities for businesses large and small to build new services, identify savings and improve operations

- open data is opening up new opportunities for businesses to connect with customers

  - Transport for London (https://tfl.gov.uk/info-for/open-data-users/) has released open data that developers have used to build over 800 transport apps

  - Thomson Reuters is using open data to connect with existing customers in order to provide better services (https://innovation.thomsonreuters.com/en/labs/data-identifiers.html)

Unlocking value from Open Data - 3

Opportunities for government: government is particularly significant in collecting broad range of different types of data

- the quantity and centrality of the data it collects
- most of that government data is public data by law

Areas where open data is creating value for government:

- Transparency and democratic control: open data can help make governments more transparent. It can provide the evidence that public money is being well spent and policies are being implemented
    - According to David Eaves, open data allowed citizens in Canada to save the government $3.2bn in fraudulent charitable donations in 2010 (https://eaves.ca/2010/04/14/case-study-open-data-and-the-public-purse/)
    - In Nigeria, Follow the Money (https://followthemoneyng.org/) ensures that public funds are spent implementing the policies promised to the people

Unlocking value from Open Data - 4

- Economically: several studies have estimated the economic value of open data at several tens of billions of Euros annually in the EU alone
  - The Danish husetsweb.dk helps you to find ways of improving the energy efficiency of your home, including financial planning and finding builders who can do the work
  - Google Translate uses the enormous volume of EU documents that appear in all European languages to train the translation algorithms, thus improving its quality of service

- Increase government efficiency:
  - The Dutch Ministry of Education has published all of their education-related data online for re-use and the number of questions they receive has dropped

- Making government more effective:
  - The Dutch department for cultural heritage is actively releasing their data and collaborating with amateur historical societies and groups such as the Wikimedia Foundation in order to execute their own tasks more effectively

Unlocking value from Open Data - 5

- Improved service delivery: governments need to balance the demands of growing populations with the need to tackle small-scale, local issues
  - The availability of detailed open data is essential to improving delivery of services at the local level
  - mySociety (https://www.mysociety.org/) helps people be active citizens with technology, research and data that individuals, journalists, and civil society can use, openly and for free

- Cost savings: open data allows governments to make savings in key areas such as healthcare, education and utilities
  - In the UK, open data helped reveal £200 million of savings in the health service
  - In France, energy data is being used to drive more efficient energy generation practices

Unlocking value from Open Data - 6

- Saving lives: open geographic data and aid statistics are being used by humanitarian groups to deliver targeted supplies in disaster zones
  - Open mapping data helped disaster response teams target aid delivery during the 2010 Haiti earthquake (https://www.hotosm.org/projects/haiti_)
  - Open data was also used for responses to the Philippines typhoon in 2014

- Culture: open data is connecting people with important cultural issues and helping to shape a more informed debate around them
  - OpenGLAM (https://openglam.org/) is helping to capture the heritage and cultural memories of groups in Germany, Switzerland and Finland.

Unlocking value from Open Data - 7

- Benefiting culture and the environment: open data is helping people target initiatives for cultural and environmental benefits

- Environment: open data helps farmers to improve yields and support a growing population without the need to destroy valuable habitats
  - Plantwise (https://www.plantwise.org/) are collecting open data to produce valuable information packs for farmers about plant health and threats from diseases
  - GODAN (Global Open Data for Agriculture and Nutrition) (https://www.godan.info/) is a rapidly growing group with partners from national governments, non-governmental, international and private sector organisations aimed at implementing innovations that help achieve the UN Sustainable Development Goal 2 - ending global hunger, achieving food security, improving nutrition and sustainable agriculture by 2030

Open data: agent of change - 1

- The most successful open data initiatives share similar characteristics
- Understanding successful approaches can help unlock the value of open data
- Implementing an open data initiative often involves cultural and institutional change
- Opening data goes far beyond putting data on a website under an open licence
  - Applying the technology is the easy part – bringing about cultural change can be much harder

Open data: agent of change - 2

Leadership and engagement: active leadership and engagement helps foster successful open data initiatives and bring a positive change

- As an open data program is launched, it is common for those driving it to meet some resistance from within their organisations
  - Active support from the top of the organizations can help
- Engagement with civil society, business and government are key aspects of any open data initiative
  - Successful initiatives tend to start with an open dialogue between open data publishers and consumers
- Strong leadership and active engagement with key stakeholders is key to any successful initiative.
  - Active leadership delivers a 'push' from the organisation, active dialogue creates a 'pull' from data users

Open data: agent of change - 3

Supply and demand: when examining an open data initiative, consider the following aspects:

- **Your approach:** as open data literacy grows, people are starting to think first about the needs of data consumers and shape their initiatives accordingly
  - a staged approach helps to ensure that an organisation achieves change in small steps

- **Who wants your data:** There is little point in publishing open data if there is no one prepared or motivated to create new value from it
  - focus on demand that currently exists, sse existing consumer dialogues, opinion surveys to see where the demand is, and follow it

- **Who will use and support your data:** a successful open data initiative has an engaged community who actively use the data, with access to resources that support them
  - a strong community of open data reusers has a sense of ownership over the data that cover both the data and the outputs generated from it
  - it is important for publishers and consumers to fully understand each others' perspectives
  - a strong open data community is the Open Street Map project (https://www.openstreetmap.org/)

Open data: agent of change - 4

Culture change for new markets: applying technology is the easy part. Implementing culture change can be much harder

- **Identifying new markets:** the traditional market for open data, where governments publish data for others to consume, has changed
  - Governments, businesses and society are all now both suppliers and consumers of open data

- **Culture change:** adopting open data will often require a change in the way an institution operates
  - people driving initiatives need to spend time developing a shared vision, overcoming barriers and building coalitions for change

- **Planning for change:** ways to handle and plan for culture change are becoming clearer.
  - Governments, organisations and individuals should have free access to key education, tools and guides that help plan for the future
  - Pilots and practical initiatives play important roles in understanding the impact of open data and the change process

What can open data do for you?

https://vimeo.com/110800848

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